

Nishant Hegde

Rochester, New York | +1 (585)-559-7746 | nishant.hegde13@gmail.com | [LinkedIn](#) | [GitHub](#) | [Portfolio](#)

Education

University of Rochester

M.S. in Data Science

Aug 2024 to Dec 2025

Rochester, New York

Work Experience

Data Science Intern | EagleHawk One, Inc

Buffalo, New York (Remote)

Jun 2025 to Present

- Led the development of 3 MVP deep learning solutions for the company's AI platform, covering object detection, instance segmentation, and retrieval-augmented generation (RAG) use cases.
- Designed an object detection pipeline using YOLOv8 and PyTorch to detect streetlights in nighttime drone imagery, implementing data processing with Pandas and NumPy and processing 5,000+ images with 91% detection accuracy.
- Engineered a thermal imaging-based rooftop leak detection system using Mask R-CNN on aerial FLIR imagery, leveraging transfer learning and AWS services (S3, SageMaker, MTurk) for scalable multi-class instance segmentation.
- Built an offline hybrid RAG system in Python to optimize gas facility safety training, using NLP and Scikit-learn for document processing, integrating Chonkie, Docling, and Ollama LLM, and reducing training session time by 70%.

Machine Learning Intern | Laboratory for Laser Energetics

Rochester, New York

Aug 2025 to Dec 2025

- Applied deep neural networks to model laser physics, predicting laser beam behavior through the OMEGA EP amplification system to advance understanding of high-energy laser systems.
- Designed a U-Shaped Fourier Neural Operator (U-FNO) and a 3D Attention Recurrent Residual U-Net to jointly predict laser spatial beam profiles and temporal pulse shapes, achieving 1% prediction error across spatio-temporal domains.
- Engineered HPC training infrastructure with model parallelism, mixed-precision training, and gradient checkpointing across A100/H100 clusters, reducing training time by 40% and GPU VRAM usage by 30%.
- Created and presented data visualizations using Python (Seaborn, NumPy, Matplotlib, Pandas) to convey insights to non-technical, cross-functional team members and align project goals.

Software Engineer | Toshiba (KIOXIA America, Inc)

Folsom, California (Remote)

Aug 2020 to May 2024

- Designed and validated firmware for enterprise NVMe SSDs to ensure reliability and performance in data center applications.
- Developed regression, stress testing, and failure analysis test suites in Java that improved firmware stability by 15%.
- Orchestrated the deployment of high-quality firmware modules by using Scrum and Agile methodologies.
- Automated pipelines with Jenkins, enabling seamless nightly builds causing a 20% reduction in manual QA effort.
- Streamlined collaboration through Confluence, improving documentation transparency and workflow efficiency by 33%.
- Built internal debugging and telemetry analysis tools in Python to streamline firmware failure root-cause analysis.

Projects

Wafer Map Failure Pattern Detection [[view project](#)]

- Implemented Vision Transformer (ViT) architecture for semiconductor wafer defect classification, achieving 96.2% accuracy on 170K samples through fine-tuning and data augmentation techniques.
- Engineered feature extraction and dimensionality reduction pipeline with 40+ visualizations including PCA/t-SNE clustering and radial defect pattern analysis for manufacturing insights.

NVMe RAG System [[view project](#)]

- Architected retrieval-augmented generation system for NVMe specification querying using ChromaDB vector database, sentence-transformers embeddings, and Ollama LLM, reducing manual search time by 40%.
- Implemented semantic chunking strategy, cross-encoder reranking, and document processing pipeline via LangChain and FastAPI for scalable technical knowledge retrieval.

Social Media Intelligence Engine [[view project](#)]

- Engineered streaming ML pipeline using Apache Spark Streaming and Delta Lake to process 40K+ JSON files via medallion architecture on Databricks for real-time sentiment analysis.
- Built end-to-end ETL system with Spark UDFs, structured streaming, and MLflow model registry for automated model deployment and performance monitoring at scale.

Technical Skills

Languages & Databases:

Python, R, C++, MATLAB, Java, SQL, MongoDB, PostgreSQL, Bash

ML/DL Frameworks:

PyTorch, TensorFlow, Scikit-Learn, NLTK, CUDA, Apache Spark, Databricks

Cloud/DevOps & Visualization:

Docker, Heroku, MLflow, Delta Lake, Tableau

Specializations:

Image Enhancement, Object Detection, Semantic Segmentation, Classical ML, R-CNNs

Other Tools and Services:

JIRA, Confluence, Workday, LaTeX, MS Excel, MS PowerPoint, Slack, Git